Fiber-Optic and IMU Sensors for Muscle Fatigue Detection in Work Settings

Sophia Otálora[1], Marcelo Eduardo Vieira Segatto[1], Maxwell E. Monteiro[2], Carlos A. Cifuentes[3], and Camilo A. R. Diaz[1]

***Abstract*—Detection of muscle fatigue is essential in industrial environments to prevent work-related disorders and improve worker safety. Also, measuring elbow position is essential in human motion analysis. This paper proposes a novel approach for detecting muscle fatigue using an Optical Fiber (OF) sensor and Inertial Measurement Unit sensors (IMU). Ten healthy subjects performed a repetitive lifting activity using the IMU and OF sensor until reaching muscle fatigue. The study evaluates four machine learning algorithms, including Ensemble Classifiers (EC), Support Vector Machine (SVM), k Nearest Neighbour (k-NN), and Neural Networks (NN). Results of detecting muscle fatigue using the OF and IMU sensors obtained an accuracy of 98.4%. This setup can be easily used by workers and be relevant to identify workers' fatigue during long working shifts.**

***Index Terms*—Fatigue detection, Optical fiber sensor, Inertial sensor.**

I. Introduction

Around 1.7 billion people worldwide suffer from musculoskeletal disorders (MSDs), including low back and neck pain [1]. Many of these disorders are lifelong illnesses with consequences such as long-term pain, physical disability, loss of independence, decreased social interaction, and reduced quality of life [2]. The MSDs can be caused when repetitive load-handling activities are performed, resulting in muscle fatigue, motor impairment, and kinematic variability. According to the Global Burden of Disease Study, this disease is the leading cause of years lost to disability, and it is estimated to increase over the upcoming years [3].

Muscle fatigue is known as the inability to maintain a desired power output, resulting in a muscular contraction under a load [4]. Detection of muscle fatigue in industrial environments is essential to prevent injuries and improve workers' safety. There are studies for estimating muscle fatigue with methods such as joint angle alterations in repetitive tasks, detecting angular changes at the trunk, shoulder, and elbow [5], and also with Inertial Measurement Units (IMUs) or heart rate sensors identifying localized fatigue at the back [6].

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[1]S. Otálora, M. E. V. Segatto and, C. A. R. Diaz are with Graduate Program of Electrical Engineering, Federal University of Espirito Santo, Vitoria 29075-910, Brazil. sophia.gonzalez@edu.ufes.br, marcelo.segatto@ufes.br, camilo.diaz@ufes.br

[2]M. Monteiro is with Instituto Federal do Espírito Santo (IFES), Campus Serra, Serra, Brazil. maxwell.monteiro@ifes.br

[3]C. A. Cifuentes is with the Bristol Robotics Laboratory, University of the West of England, Bristol, UK. carlos.cifuentes@uwe.ac.uk

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Optical Fiber (OF) sensors are currently used due to multiple advantages due to their flexible structure, low cost, lightweight, and immunity to electromagnetic interference [7], which are important characteristics in industrial environments. These sensors have been used for angle estimation, which is essential in robotic applications to estimate elbow range of motion (ROM) and angular displacements [8]. One of the simplest principles used is the intensity variation due to ease of implementation to sense curvature, temperature, and pressure [9]. Leal et al. created a sensor for lower limb exoskeleton instrumentation. When validated with a potentiometer, the sensor demonstrated an error of 1.80° being an option for robotic instrumentation [10].

The combination of OF sensors and wearable commercial sensors represents a novel and innovative approach to detecting muscle fatigue for industrial applications. OF sensors allow the monitoring of joint angles, while IMU sensors can provide additional information about arm movement in different planes during induced fatigue. The contribution of this paper is to estimate muscle fatigue of the Biceps Brachii muscle in a repetitive lifting task through wearable sensors such as OF sensors and IMU sensors.

II. METHODOLOGY

A. Sensor implementation

A printed three-dimensional polylactic acid (PLA) structure was built for the elbow joint. The structure was secured using velcro, and an OF was threaded through the system. The polymer OF (SH4001, Mitsubishi Chemical Co.) is used to measure the angle of the elbow. A Light-emitting diode (LED) IF-E97, and a phototransistor IF-D92 (Industrial Fiber Optics, USA) located on opposite sides of the fiber are utilized to measure changes in voltage. To improve the sensor response, a sensitive zone [11], [12] is created, which increases the optical power losses leading to higher voltage changes in response to the elbow joint's movements. To acquire the data from the OF, a microcontroller Teensy 3.6 (PJRC, USA) with an analog-to-digital converter (ADC) of 16-bit resolution was used. Additionally, two Alt-IMU V4 sensors, read by I2C communication protocol, were placed on the participant's arm and forearm through the 3D structure as shown in Figure 1. Then, the angle sensor is characterised using KINOVEA software with six flexion and three extension angles.

B. Fatigue protocol

Ten healthy volunteers, including five women and five men (age: 24 ± 3 years old, height: 174 ± 2 cm, and weight: 70

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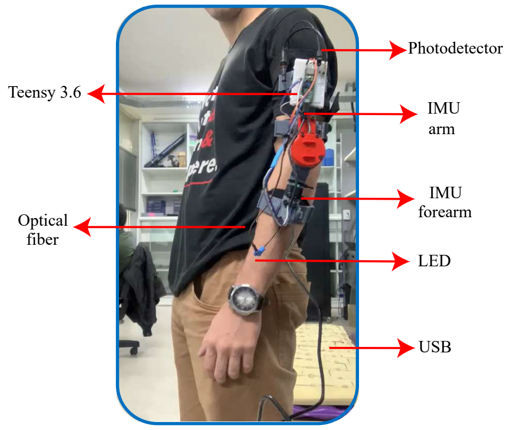

Fig. 1. Subject wearing the 3D structure with all of the electronics and sensors mounted.

± 5 kg) performed the three following stages:

- Stage 1: The test starts with a calibration phase, where the subject must stand still for 10 seconds. This allows the sensor to record a stable baseline measurement, to subtract the offset from subsequent readings.
- Stage 2: The warm-up phase prepares the participant's muscles and joints for the fatiguing exercises. During this phase, the participant performs ten repetitions of the biceps curl using a load (1 kg for women, 2 kg for men).
- Stage 3: In the fatigue test, participants perform biceps curls with an increased load (2.5 kg for women, 4.5 kg for men) until they are no longer able to make a repetition or maintain the range from the leg to the shoulder. Previous studies have shown that increasing the load is the factor that most generates muscle fatigue in repetitive weight lifting in the biceps muscle [13]. Also, the Borg scale was used to label the user's fatigue condition. The scale is from 0 to 10 and was asked to the volunteer every 20 seconds during the activity [14].

C. *Data processing and feature extraction*

MATLAB software was used for data processing. All signals were acquired at a sampling frequency of 52 Hz. For the OF signal, a low-pass filter with a cut-off frequency of 0.5 Hz was used to reduce noise. This value was chosen by applying the fast Fourier transform. After this, the offset is removed and the signal is divided according to the Borg scale into three fatigue states: low (1-3), moderate (4-7), and high (8-10). Then, the signal is divided into windows by inter-subject repetition for feature extraction and to divide all IMU signals according to the repetition. For the IMU signal, a 30 ms moving window filter is used and the signals are divided according to the repetitions set in the OF signal for feature extraction. Finally, 4 features were extracted from the IMU sensor (Gyr and Acc) including mean, standard deviation (STD), Root Mean Squared (RMS), and the amplitude of the signal. And for the OF sensor, the time to perform each repetition was added, which was normalized to the one with the shortest inter-subject time.

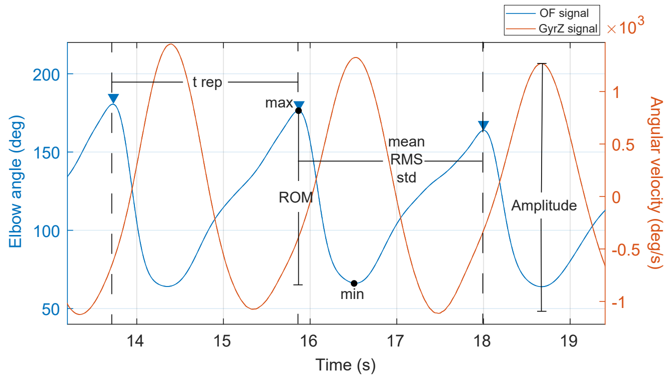

Fig. 2. Features extracted of OF and GyrZ elbow signals for two repetitions in the moderate state of fatigue.

TABLE I
TEST ACCURACY'S WITH THE WEARABLE SENSORS FOR THE 4 MODELS.

Model	Type	Accuracy w/o PCA (%)	Accuracy with PCA (%)
EC	Subspace kNN	98.4	98.4
SVM	Fine Gaussian	91.3	96.8
kNN	Fine	96.8	95.2
NN	Medium	96.0	94.4

These features are best shown in Figure 2. We computed 24 features for each IMU in total (48 features) and 5 features from the OF sensor for a total of 53 features.

Finally, the information was divided into 80% for training and 20% for testing the states' fatigue detection. In addition, k-fold cross-validation is used with k=8 corresponding to the number of volunteers [15]. For algorithm classification, the MATLAB Classification Learner tool was used. Four algorithms that have been used for fatigue identification were used to recognize the states: Ensemble Classifiers (EC), Support Vector Machine (SVM), k Nearest Neighbour (k-NN), and Neural Networks (NN) [16]. For the algorithm's performance, metrics such as accuracy, precision, recall, and F-score were used. Lastly, the Principal component analysis (PCA) was used for feature reduction.

III. RESULTS AND DISCUSSION

A. *Feature reduction*

After using the PCA method during training for feature reduction using 99% of the variance, 10 components were kept, resulting in 10 features out of 53. The results with these characteristics can be observed in Table I, where the high-performing algorithm (Ensemble type subspace k-NN) maintained the accuracy after feature reduction with a 98.4%.

B. *Fatigue classification*

The confusion matrix when classifying muscle fatigue with the EC is presented in Figure 3. It is observed that the most difficult class to classify is moderate fatigue which is often confused with high fatigue. This occurs when the person is reaching the end of the moderate stage and begins to have the characteristics of the fatigue state such as decreased range of

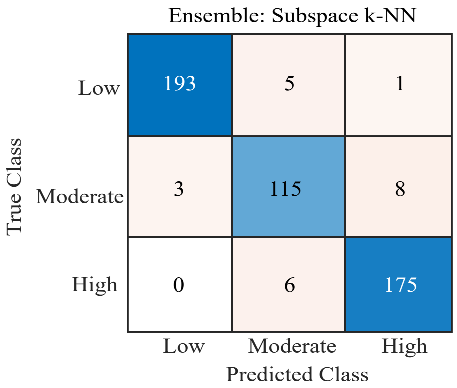

Fig. 3. Ensemble Algorithm Confusion Matrix for 10 features.

motion and velocities. Results of the EC algorithm include an accuracy of 98.4%, a precision of 94.9%, a recall of 94.8%, and an F-score of 94.6%. This algorithm has greater performance than a study where they estimate biceps muscle fatigue using EMG signals where SVM obtained an accuracy of 96.5% [17]. However, they do not use wearable sensors for the estimation. In comparison to a study where they use IMU and heart rate sensors, where they obtain an accuracy of 88% with a feedforward NN for the cross-subject model [15]. IMU sensors for fatigue assessment are used to observe changes in the movement patterns for an action. In addition, it quantifies kinematic parameters being comparable when the person is not in a fatigued state [18]. In our study, this is relevant because when individuals experience localized fatigue, their movement velocities decrease, changing the horizontal axes as a result of compensatory strategies.

Although most of the studies for biceps brachii fatigue detection are based on EMG analysis using different methods such as Polynomial Chirplet Transform [19] or Discrete Wavelet Transform [20], it is important to emphasize that for industrial environments, it should be sought to have the least amount of sensors to avoid worker discomfort and data loss due to the prolonged use of these sensors in demanding activities. Therefore, low-cost and wearable sensors are used to detect muscle fatigue in order to apply it in future works providing feedback to the worker when reaching a fatigued state to control the activity being performed, and it can also be connected to assistive devices to improve their control strategies.

IV. Conclusions and Future Work

In conclusion, an algorithm for the detection of 3 states of muscle fatigue was obtained with an accuracy of 98%, being a high performance compared to the literature. The detection by means of wearable sensors allows it to be non-invasive, low-cost, and lightweight, which makes it an essential tool in robotic applications where commercial sensors may have this limitation. Future work is expected to implement the study with a larger sample and validate the model with EMG sensors to be applied in user fatigue prevention by generating feedback and in the control strategy of assistive devices.

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